

CS220 Design and Analysis of Algorithms

School of Computer Science and Engineering

Macau University of Science and Technology

Assignment 1

Due Day: 12 March, 2026 (Thursday)

1. Use the **substitution method** to show that each of the following recurrences defined on the reals has the asymptotic solution specified. **You have to show both the upper (O) and lower (Ω) bounds of $T(n)$.**
 - a. $T(n) = 2T(n/3) + \theta(n)$ has solution $T(n) = \theta(n)$.
 - b. $T(n) = 4T(n/2) + \theta(n)$ has solution $T(n) = \theta(n^2)$.
2. For each of the following recurrences, sketch its **recursion tree**, and guess a good asymptotic **upper bound** on its solution. **YOU DO NOT need to use the substitution method to verify your answer.**
 - a. $T(n) = T(n/5) + \theta(n^3)$.
 - b. $T(n) = 5T(n/2) + \theta(n)$.
3. Use the **master method** to give **tight asymptotic bounds** for the following recurrences:
 - a. $T(n) = 3T(n/9) + \sqrt{n} \lg^2 n$.
 - b. $T(n) = 2T(n/4) + n^2$.